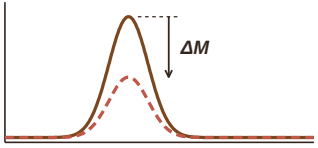
## a Four spatial failures, one RMSE (0.17)

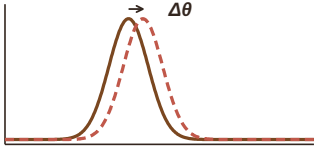
reference — estimate

### Magnitude



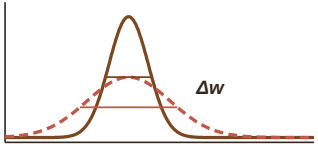

peak 1.00 → 0.50
**RMSE 0.17**

### Location



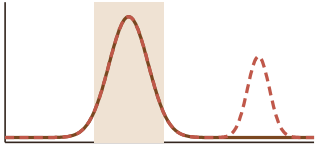

centroid +0.05 of the wall
**RMSE 0.17**

### Extent




FWHM 0.15 → 0.32
**RMSE 0.17**

### Support

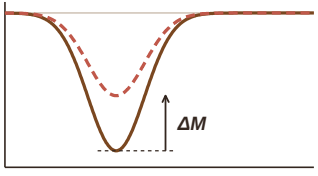


extra peak 0.67 off the support
**RMSE 0.17**

position around the wall →

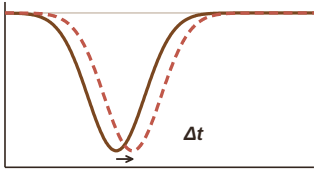
## b Temporal: same RMSE (0.16)

### Magnitude




peak −1.00 → −0.60
**RMSE 0.16**

### Timing



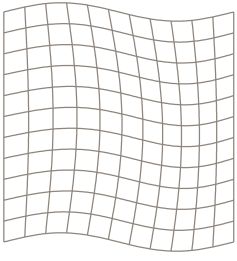

peak time +0.05 cycle
**RMSE 0.16**

time →

A single mean-squared error cannot say which attribute failed; report bias, location, extent and timing error separately.

## c Invertible is necessary, not sufficient

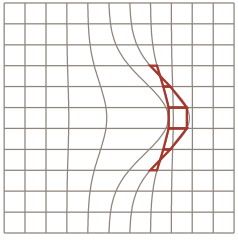
### Invertible



| | |
|---|---|
| min det J | **0.81** |
| cells det J ≤ 0 | **0 / 121** |

can still be wrong

### Folded



| | |
|---|---|
| min det J | **−1.13** |
| cells det J ≤ 0 | **5 / 121** |

orientation reversed